

LITERATURE SURVEY

Autonomous AI Agent for IT Helpdesk

Course	CSS7102 – Mini Project
Review	Review 1 (CA-01)
Problem Statement	PSAIAC_60

1. Purpose

This literature survey reviews research relevant to an Autonomous AI Agent for IT Helpdesk. The selected papers cover machine-learning-based helpdesk ticket classification, machine learning for IT service management, service-request and incident workflows, LLM-based autonomous agents, and privacy in tool-using LLM agents. Together they provide the domain and technical background for the proposed system.

2. Literature Review

[1] A machine learning based help desk system for IT service management

Research Summary: F. Al-Hawari and H. Barham present a web-based help desk and a machine-learning ticket-classification model that associates tickets with the correct service. Including ticket comments and descriptions improved reported prediction accuracy from 53.8% to 81.4%. The system also supports service requests, ticket management, notifications and reporting.

Relevance: Directly supports AI-assisted helpdesk classification and automation. It provides a domain foundation for understanding and routing L1 requests.

Limitation / Opportunity: It mainly classifies and manages tickets; it does not implement a modern LLM agent that reasons and executes multi-step tools.

Publication: Open access; DOI 10.1016/j.jksuci.2019.04.001.

[2] Machine Learning in IT Service Management

Research Summary: D. Zuev, A. Kalistratov and A. Zuev propose a machine-learning model for predicting infrastructure incident resolution time. The paper treats resolution time as an important ITSM performance indicator and investigates ML to improve handling efficiency and reduce service costs.

Relevance: Supports data-driven improvement of ITSM and provides a basis for reducing service-desk effort.

Limitation / Opportunity: It predicts resolution time rather than autonomously resolving user requests and does not combine LLM reasoning, knowledge retrieval and tool execution.

Publication: Open access; DOI 10.1016/j.procs.2018.11.063.

[3] Analysis of COBIT 5 Process “DSS02 - Manage Service Requests and Incidents” for the Service Desk Using Process Mining

Research Summary: I. Tello, C. Ruiz and S. G. Yoo analyze incident management in a real oil-service company using process mining. The study considers classification, recording, prioritization, fulfillment flow and event states to identify recurring events and process shortcomings.

Relevance: Provides a workflow foundation for classifying, prioritizing and fulfilling service requests and incidents.

Limitation / Opportunity: It analyzes existing processes rather than implementing an autonomous AI agent for natural-language understanding and authorized action execution.

Publication: IEEE conference paper; DOI 10.1109/ICEDEG.2018.8372335.

[4] A survey on large language model based autonomous agents

Research Summary: L. Wang and coauthors provide a comprehensive survey of LLM-based autonomous agents, covering agent construction, a unified framework, applications, evaluation, challenges and future directions. The framework discusses components such as profile, memory, planning and action.

Relevance: Provides the core AI-agent foundation for an agent that understands requests, plans actions and interacts with an environment.

Limitation / Opportunity: It is a broad survey rather than an IT-helpdesk implementation, so its concepts must be adapted to controlled L1 workflows.

[5] PrivacyAsst: Safeguarding User Privacy in Tool-Using Large Language Model Agents

Research Summary: X. Zhang and coauthors study privacy risks when LLM agents interact with auxiliary tools. The paper identifies challenges involving privacy requirements, generalization across LLMs and tools, and unrestricted tasks, and presents PrivacyAsst.

Relevance: Relevant because an IT helpdesk agent may use authorized tools or APIs. It supports privacy safeguards, authorization and controlled tool execution.

Limitation / Opportunity: It focuses on privacy for tool-using LLM agents rather than the complete IT helpdesk lifecycle of classification, knowledge retrieval, workflow execution, verification and escalation.

Publication: IEEE Transactions on Dependable and Secure Computing; DOI 10.1109/TDSC.2024.3372777.

3. Comparative Analysis

Area	Existing research	Project implication
Helpdesk automation	ML can classify tickets and support helpdesk management.	AI-based understanding and routing of L1 requests.
ITSM prediction	ML can improve incident-handling efficiency.	Use data-driven evaluation of workflows.
Service workflows	Requests/incidents can be classified, prioritized and fulfilled.	Represent approved L1 tasks as controlled workflows.
Autonomous agents	LLMs support planning, reasoning and action.	Use an agent architecture for autonomous task handling.
Tool security	Tool-using agents create privacy/security concerns.	Use authorization, controlled tools and escalation.

4. Research Gap

The literature covers important components separately: AI-assisted helpdesk classification, machine learning for ITSM, service-request and incident workflows, LLM-based autonomous agents, and privacy for tool-using agents. A project-level opportunity remains to integrate these capabilities into a controlled autonomous L1 IT helpdesk agent that understands a user request, retrieves relevant organizational knowledge, selects an authorized workflow, performs an allowed tool action, verifies the outcome, and escalates cases that cannot safely be completed. Initial target workflows are password reset, software installation and access request.

5. Conclusion

The literature supports combining AI-based helpdesk processing with modern LLM-agent capabilities. The proposed project focuses on integrating these capabilities into a practical, controlled IT helpdesk workflow.

6. References — IEEE Style

[1] F. Al-Hawari and H. Barham, "A machine learning based help desk system for IT service management," Journal of King Saud University – Computer and Information Sciences, vol. 33, no. 6, pp. 702–718, 2021, doi: 10.1016/j.jksuci.2019.04.001.

[2] D. Zuev, A. Kalistratov, and A. Zuev, "Machine Learning in IT Service Management," Procedia Computer Science, vol. 145, pp. 675–679, 2018, doi: 10.1016/j.procs.2018.11.063.

[3] I. Tello, C. Ruiz, and S. G. Yoo, "Analysis of COBIT 5 Process 'DSS02 - Manage Service Requests and Incidents' for the Service Desk Using Process Mining," in 2018 Fifth International Conference on eDemocracy & eGovernment (ICEDEG), 2018, pp. 304–310, doi: 10.1109/ICEDEG.2018.8372335.

[4] L. Wang, C. Ma, X. Feng, Z. Zhang, H. Yang, J. Zhang, Z. Chen, J. Tang, X. Chen, Y. Lin, W. X. Zhao, Z. Wei, and J. Wen, "A survey on large language model based autonomous agents," Frontiers of Computer Science, vol. 18, no. 6, Art. no. 186345, 2024, doi: 10.1007/s11704-024-40231-1.

[5] X. Zhang, H. Xu, Z. Ba, Z. Wang, Y. Hong, J. Liu, Z. Qin, and K. Ren, "PrivacyAsst: Safeguarding User Privacy in Tool-Using Large Language Model Agents," IEEE Transactions on Dependable and Secure Computing, vol. 21, no. 6, pp. 5242–5258, 2024, doi: 10.1109/TDSC.2024.3372777.

Source note: Paper descriptions are based on the papers' available published content and metadata. The comparative analysis, research gap and project implications are interpretations for this Review-1 project.